

Notes on sensor array nearest neighbor interaction theory project

Resonant Coupled Oscillator Simulator
Physics, Optimization, and Results

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August 26, 2026

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1. Overview

ResOSc is a Python package for simulating and analyzing chains of coupled mechanical oscillators, with direct application to precision physics experiments such as gravitational-wave (GW) detectors and dark-matter (DM) sensors. Given a set of masses m_i , wall springs k_{ii} , and coupling springs $k_{i,i+1}$, the package:

1. solves the eigenvalue problem to obtain normal-mode frequencies ω_i^* and eigenvectors;
2. computes the forced response under a chosen physical driving model;
3. finds the optimal linear observable (weight vector over oscillator readouts) that maximizes worst-case detection sensitivity; and
4. searches the full parameter space via Monte Carlo to find configurations that simultaneously maximize sensitivity and frequency coverage.

1.1 Roadmap: From Dimensionless Theory to a Physical Detector Forecast

The results in these notes were first developed in dimensionless units with an amplitude-based figure of merit. The programme now underway grounds the pipeline in the physical framework of the companion note *SNR of a Multimode Mechanical Detector* and the levitated-sensor detector (LSD) platform [Aggarwal *et al.*, PRL **128**, 111101 (2022); arXiv:2010.13157], targeting the science case of planetary-mass primordial black hole (PBH) binaries [Miller *et al.*, PRL **133**, 111401 (2024)] and weakly interacting (“gentle”) dark matter. The plan:

1. **Objective.** Replace the amplitude minimax by the matched-filter signal-to-noise ratio $\rho^2 = 4 \int |\tilde{h}(f)|^2 / S_h(f) df$, reported as a *horizon distance* $d_{\max}(\mathcal{M}_c)$ for benchmark PBH chirp masses (inspiral spectrum $|\tilde{h}|^2 = \mathcal{A}^2 f^{-7/3}$), and as minimum detectable impulse / coupling for the dark-matter channels. The full transfer-function integral treats overlapping resonances exactly, retiring the isolated-peak/swept metric dichotomy of Sec. 6; the resolvability condition survives only as the validity criterion of the per-mode approximation. The broadband source spectrum also subsumes the former ad-hoc frequency-spread objective.
2. **Noise model.** Thermal force noise from the fluctuation–dissipation theorem ($S_F = 4k_B T m_i \gamma_i$, uncorrelated in UHV) plus readout shot noise, combined into the strain-referred PSD $S_h(f) = S_O^{\text{noise}}(f) / |T(f)|^2$.
3. **Two readouts, compared.** Readout A (axial cavity probe: one channel, weights fixed by the intracavity field — design freedom is purely mechanical) and Readout B (transverse imaging: N channels, freely optimizable weights and probe-power allocation). Both are computed for every design; thermal- and readout-limited regimes are both evaluated, with the crossover frequency reported.
4. **Units and benchmark.** SI throughout; the benchmark parameter block follows the LSD proposal (stacked SiO₂/Si discs, $r = 75 \mu\text{m}$, trap frequencies 10–100 kHz, $P = 10^{-11}$ Torr, $L = 10$ m cavity, damping 0.17–1.7 Hz, target $\sqrt{S_h} \sim 10^{-21}$ – 10^{-22} Hz^{−1/2}), with unconfirmed entries (disc mass, detected powers, quantum efficiency, cavity pole, effective temperature) flagged for confirmation.
5. **Physical coupling constraints.** Couplings are parameterized by mechanism rather than free box bounds; the first implementation uses Coulomb coupling between charged particles ($k_{ij} \propto q_i q_j / d_{ij}^3$, long-range tails included), whose stray-charge floor gives the coupling-floor constraint of the route-3 study its physical origin. Optical-binding (correlated noise) and

cavity-mediated (non-symmetric stiffness) couplings are documented extensions.

6. **Deliverable.** Sensitivity curves $\sqrt{S_h(f)}$ and PBH horizon distances for optimized coupled arrays versus the best uncoupled array, under laboratory parameters — restating the coupled-vs-uncoupled questions of these notes in units a lab can act on.

Table 1: The two readout strategies (companion note Sec. 8) and their consequences for the optimization pipeline. Both observe the same resonant motion of the coupled array; they differ in what may be optimized and in the readout-noise shape.

	Readout A (cavity)	Readout B (imaging)
Geometry	axial cavity probe, $\lambda = 1064$ nm	transverse imaging, $\lambda = 532$ nm
Channels	1 (one optical phase)	N (one per particle)
Weights w_i	<i>fixed</i> by the intracavity field, $\mathbf{w} = \mathbf{g}/ \mathbf{g} $	free post-processing choice
Readout noise	white phase shot noise; rises above the cavity pole as $1 + (f/f_{\text{cav}})^2$	flat per-particle shot noise S_i^{ro} ; probe-power allocation P_i^{sc} is an extra design knob
Pipeline freedom	mechanics only: mode shapes must be engineered to align with $\mathbf{g}$ (a mode with $N_n = 0$ is invisible)	mechanics + weights ($w_i^* \propto V_{in}/S_i^{\text{ro}}$ per mode) + power allocation

Parameters requiring confirmation. Absolute forecasts (horizon distances, $\sqrt{S_h}$ levels) are currently limited by six entries: (i) the **disc mass** (our $0.58 \mu\text{g}$ is a geometric estimate from the published disc dimensions and bulk densities); (ii) the **effective temperature** entering the fluctuation–dissipation theorem (300 K bath vs. a feedback-cooled centre-of-mass temperature[†]); (iii) the **detected probe power** P_{det} for Readout A; (iv) the per-particle **collected power** P_i^{sc} for Readout B; (v) the **detector quantum efficiency** η_{det} ; and (vi) the **cavity decay rate** κ (equivalently the cavity pole f_{cav}) and the **intracavity coupling profile** g_i , which together fix Readout A’s noise and its fixed weights; and (vii) the **controllable charge range and stray-charge floor** for the Coulomb coupling — the former bounds the coupling design space, the latter is the physical origin of the coupling-floor constraint in the route-3 study. Since $d_{\text{max}} \propto 1/\sqrt{S_h}$, these entries move the forecasts multiplicatively; all relative (coupled vs. uncoupled) statements are unaffected.

2. System Physics

2.1 Equations of Motion

A chain of n oscillators obeys

$$m_i \ddot{x}_i + \gamma_i \dot{x}_i + k_{ii} x_i + \sum_{j \neq i} k_{ij} (x_i - x_j) = f_i(t), \quad (1)$$

where γ_i is the damping coefficient (taken uniform, $\gamma_i = \gamma$ throughout), and $f_i(t)$ is the external driving force on oscillator i .

Table 2: Benchmark platform parameters implemented in `Res0Sc/physical.py` (LSD_BENCHMARK). Status: *published* = LSD proposal [PRL **128**, 111101]; *note* = companion SNR note; *estimated* = derived here; *TBC* = placeholder awaiting confirmation.

Parameter	Symbol	Value	Status
Trap frequency	f_{trap}	10 / 100 kHz cases	published
Gas pressure	P	10^{-11} Torr	published
Temperature	T	300 K (4 K opt.)	published [†]
Cavity baseline	L	10 m	published
Damping linewidth	γ	0.17–1.7 Hz + 0.005–0.05 Hz recoil	published
Disc geometry	—	$r = 75 \mu\text{m}$; SiO ₂ $14.58 \mu\text{m}$ + Si 110 nm	published
Probe wavelengths	$\lambda_{\text{probe}}, \lambda_{\text{ro}}$	1064 / 532 nm	note
Imaging aperture	NA	0.1	note
Particle spacing	d	$15.5 \mu\text{m}$	note
Disc mass	m	5.8×10^{-10} kg	estimated
Coupling mechanism	k_{ij}	Coulomb: $2q_i q_j / (4\pi\epsilon_0 d_{ij}^3)$	design variable
Achievable charge	q_i	range TBC (units of e)	TBC
Stray-charge floor	q_{min}	TBC (sets minimum $ k_{ij} $)	TBC
Detected power (A)	P_{det}	1 mW	TBC
Collected power (B)	P_i^{sc}	1 μW	TBC
Quantum efficiency	η_{det}	0.8	TBC
Cavity decay rate	κ	$2\pi \times 1$ MHz	TBC
Intracavity profile	g_i	uniform placeholder	TBC

2.2 Dynamical Matrix and Normal Modes

Dividing through by m_i and defining the *dynamical matrix* H ,

$$H_{ii} = \frac{k_{ii}}{m_i} + \sum_{j \neq i} \frac{k_{ij}}{m_i}, \quad H_{ij} = -\frac{k_{ij}}{m_i} \quad (i \neq j), \quad (2)$$

the normal-mode squared frequencies $(\omega_i^*)^2$ are the eigenvalues of H , and the corresponding eigenvectors (rows of the matrix A) give the spatial mode shapes. In the uncoupled limit ($k_{ij} = 0$ for $i \neq j$) the eigenvalues reduce to the *bare frequencies*

$$\omega_i^{(0)} = \sqrt{k_{ii}/m_i} \quad (3)$$

and A becomes the identity. The mode shapes of the baseline system are shown as stem plots in Figure 1; the full eigenvector matrix and the corresponding oscillator–mode coupling strengths are shown as heatmaps in Figure 2.

2.3 Forced Response and Peak Sensitivity

Under harmonic driving at frequency ω the complex normal-coordinate amplitude of mode i is

$$\tilde{b}_i(\omega) = \frac{q_i}{(\omega_i^*)^2 - \omega^2 + 2i\gamma\omega_i^*\omega}, \quad (4)$$

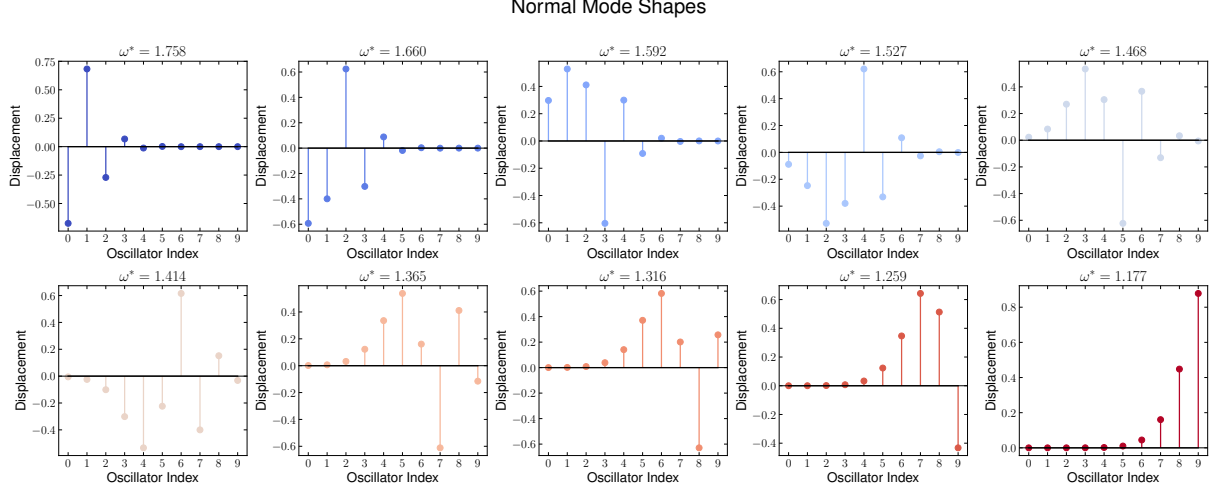


Figure 1: Normal-mode shapes of the baseline 10-oscillator system. Each panel shows the spatial displacement amplitude of one mode across all oscillators; the color maps mode index from low (blue) to high (red) frequency.

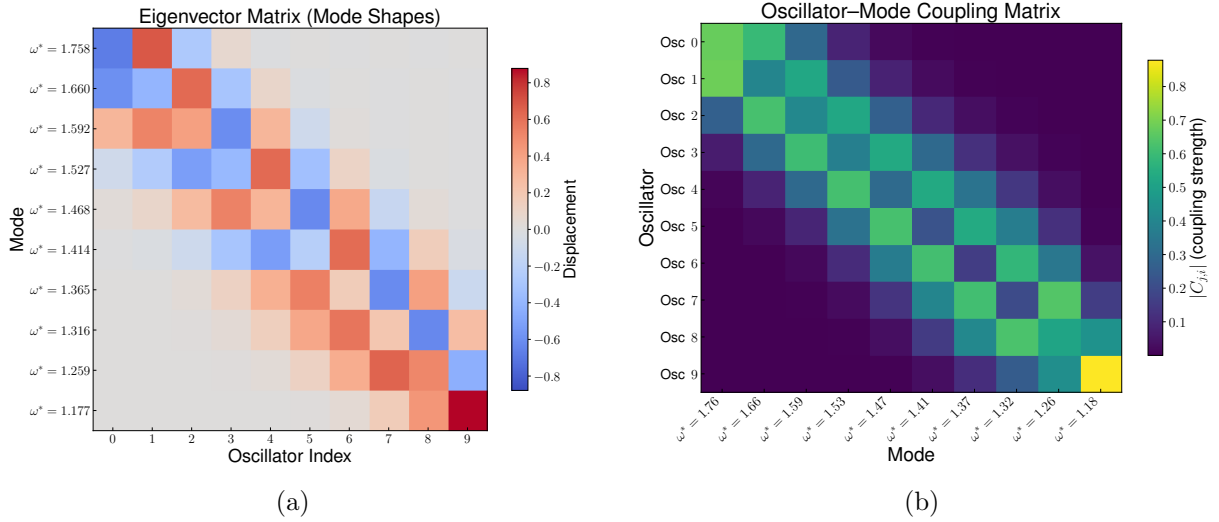


Figure 2: Heatmap representation of the normal-mode structure. (a) Eigenvector matrix A , with rows ordered by decreasing mode frequency; entry $A_{\alpha j}$ is the displacement of oscillator j in mode α . (b) Absolute coupling strengths $|C_{j,i}| = |A_{ij}|$, the transpose and absolute value of (a), showing how strongly a single-oscillator observable at position j detects mode i . The two panels carry the same information viewed from complementary perspectives.

where $q_i = \sum_j A_{ij} f_j$ is the *generalized force* on mode i . At resonance ($\omega = \omega_i^*$) the peak amplitude is

$$b_{i,\max} = \frac{|q_i|}{2\gamma(\omega_i^*)^2}. \quad (5)$$

This quantity $b_{i,\max}$ is the *peak sensitivity* of mode i : the larger it is, the more easily that mode can be detected. The forced-response amplitude spectra for the baseline system are shown in Figure 3.

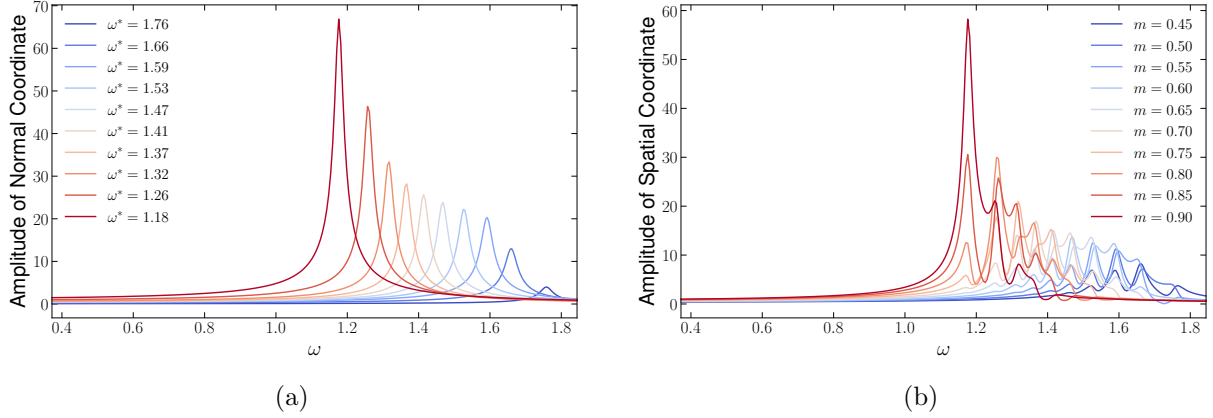


Figure 3: Forced-response amplitudes for the baseline system under GW strain driving. (a) Normal-coordinate amplitudes $|\tilde{b}_i(\omega)|$ vs. driving frequency; each peak corresponds to one normal mode and its height gives $b_{i,\max}$. (b) Spatial (oscillator-level) amplitudes vs. frequency, showing which oscillators move most at each resonance.

3. Driving Force Models

Three physically motivated driving models are implemented via the `force_model` argument to `compute_forces()`:

Model	Force on oscillator i	Physical context
strain (default)	$f_i = h_0 L k_{ii}$	GW tidal forcing, proportional to local wall stiffness
uniform	$f_i = h_0 L$	DM momentum impulse, identical kick on every mass
custom	$f_i = h_0 v_i$	Coherent DM wave or any user-defined spatial profile

The *generalized force* $q_\alpha = \sum_j A_{\alpha j} f_j$ measures how strongly the external drive projects onto each normal mode. Different spatial profiles f_j produce qualitatively different projections. For GW strain, $f_j \propto k_{jj}$, so stiffer oscillators are driven harder. For the uniform DM impulse every oscillator receives the same kick, so $q_\alpha \propto \sum_j A_{\alpha j}$: center-of-mass-like modes with large row sums are strongly driven while antisymmetric modes, for which $\sum_j A_{\alpha j} \approx 0$, are nearly undriven. The coherent DM wave modulates f_j sinusoidally, selectively exciting modes whose spatial wavelength matches the drive. The spatial profiles and their projections q_α are shown for all three models in Figure 4.

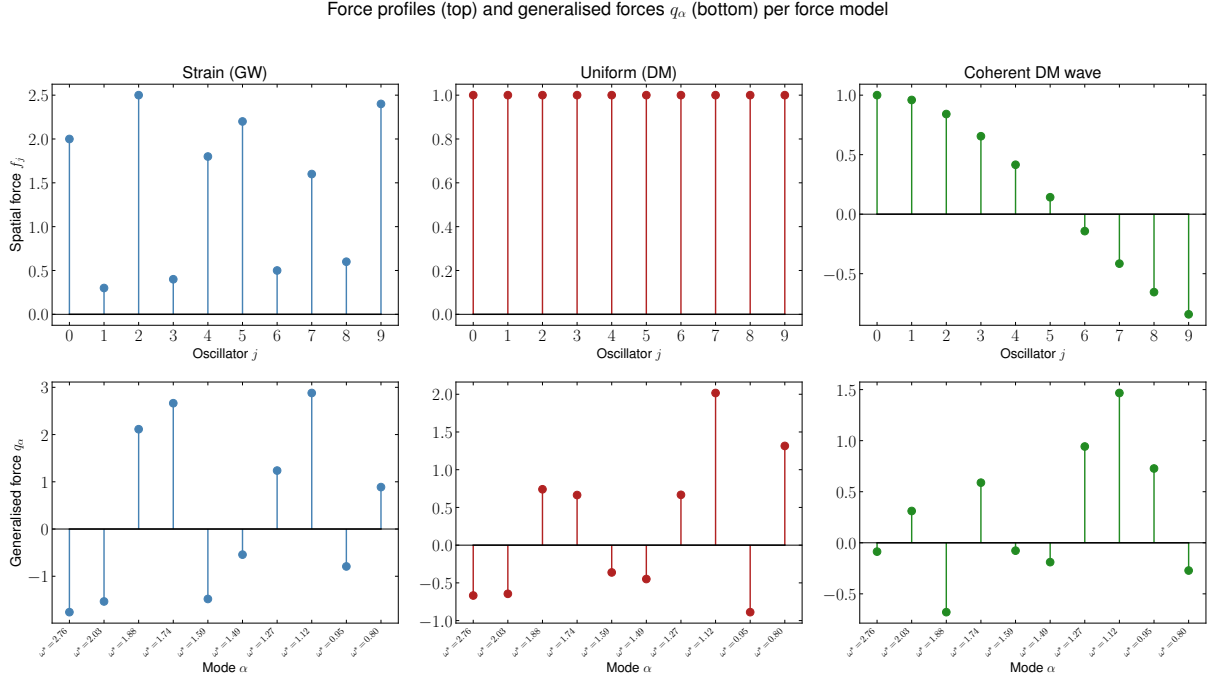


Figure 4: Spatial force profiles f_j (top row) and generalized forces q_α (bottom row) for each driving model. The GW strain force is proportional to k_{jj} ; since all wall springs are equal in the baseline system it is flat, identical in shape to the uniform impulse but rescaled by 1.2. The coherent DM wave places a half-wavelength across the chain. For flat forces, $|q_\alpha|$ grows toward the lowest mode (whose eigenvector components add most constructively) and alternates in sign between consecutive modes due to their alternating parity. The coherent wave suppresses modes whose spatial pattern is mismatched to the cosine profile.

Worst-case sensitivity by force model (baseline system). Table 3 summarizes the normalized worst-case sensitivity achieved by the optimal observable under each force model. The normalization reference is defined in Section 5; values below 1 indicate that the coupled system does not reach the uncoupled ceiling. The coherent wave comes closest because its sinusoidal envelope overlaps more uniformly with the mode shapes, avoiding the large imbalance in $|q_\alpha|$ seen under flat driving. The mechanism is made explicit in Figure 5, which decomposes q_α into the per-oscillator contributions $A_{\alpha j} f_j$, and Figure 6, which summarizes $|q_\alpha|$ and peak sensitivities across all three models.

Force model	Normalized worst-case sensitivity	Status
GW Strain	0.3495	< 1 , below reference
DM Uniform	0.3495	< 1 , below reference
DM Coherent wave	0.7874	< 1 , below reference

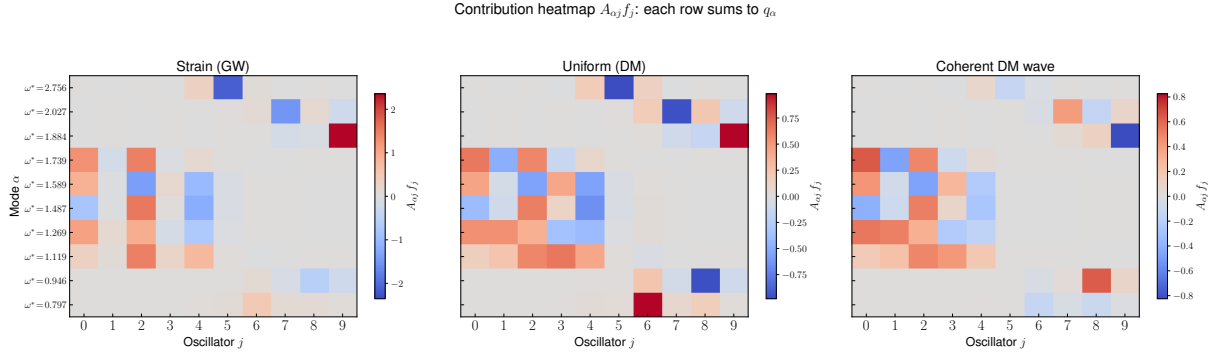


Figure 5: Contribution heatmap $A_{\alpha j} f_j$: entry (α, j) shows how much oscillator j 's force drives mode α , and each row sums to q_α . For the flat driving models the heatmap reduces to the eigenvector matrix A rescaled column-wise, so mode shapes are directly readable. For the coherent DM wave, columns near the cosine zero-crossing contribute negligibly, explaining why modes with eigenvector weight concentrated in the center of the chain nearly vanish from q_α .

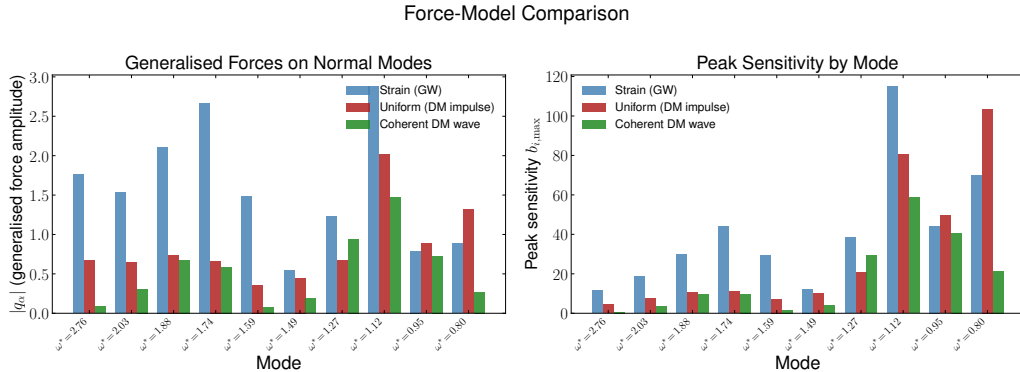


Figure 6: Per-mode generalized-force magnitudes $|q_\alpha|$ (left) and peak sensitivities $b_{\alpha, \max}$ (right) for all three driving models. The coherent DM wave produces the most balanced excitation across modes, which is reflected in its higher worst-case sensitivity in Table 3.

4. Non-Interacting Reference System and Frequency Spread

4.1 Reference System

The *non-interacting reference* is constructed from the coupled system by setting all coupling springs to zero: $k_{ij} = 0$ for $i \neq j$, while keeping masses and wall springs unchanged. Its normal modes are the individual oscillators, $A = \mathbf{I}$, with bare frequencies $\omega_i^{(0)}$.

The reference serves two purposes:

1. It provides a *frequency-spread baseline*: how much of the final bandwidth arises from varied wall springs versus coupling.
2. It provides the *sensitivity ceiling* against which the coupled system is benchmarked (Section 5).

4.2 Frequency Spread Metrics

Metric	Coupled	Reference	Difference
Absolute spread $\omega_{\max} - \omega_{\min}$	0.5816	0.4783	+0.1033
Relative spread	0.4030	0.3545	+0.0485
Spacing uniformity (CV)	0.2400	0.3021	-0.0621

Result. The coupling springs widen the frequency band by $\approx 13.7\%$ relative to the bare oscillator spread, as visible in the spacing comparison of Figure 7 and the aggregate bar chart of Figure 8. The coupled system also has more uniform mode spacing (lower coefficient of variation), which might be beneficial for broadband detection. The coupling-induced gain is the extra bandwidth that can be tuned by adjusting the coupling springs, whether in the simulation or in the lab.

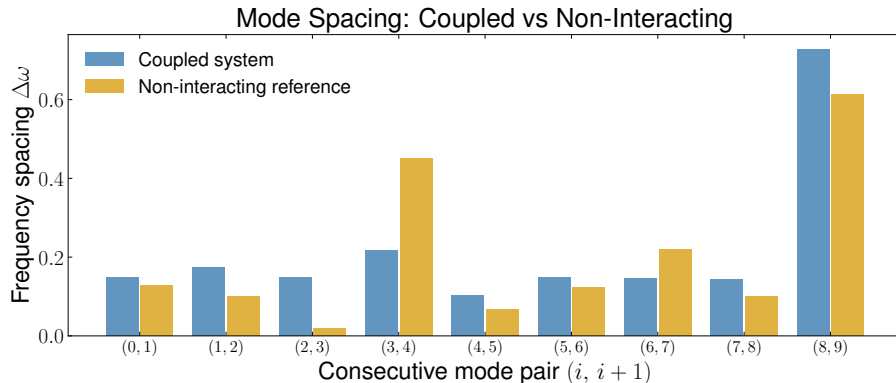


Figure 7: Consecutive mode spacings $\Delta\omega$ for the coupled system (blue) versus the non-interacting reference (yellow). Coupling pushes the modes apart unevenly, widening the total bandwidth by $\approx 13.7\%$ and reducing the spacing coefficient of variation from 0.30 to 0.24.

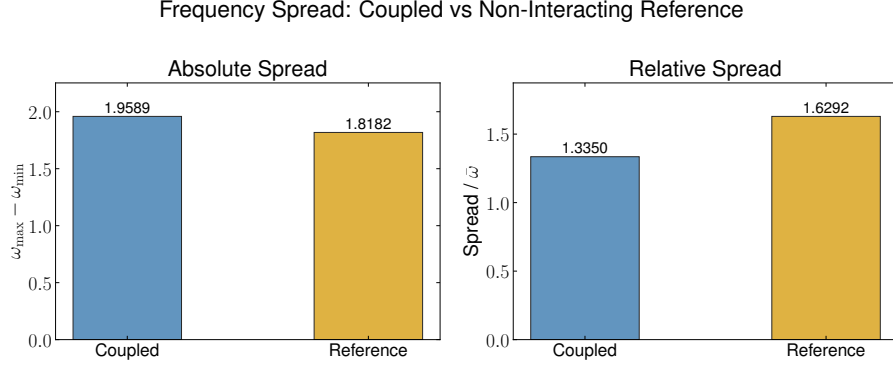


Figure 8: Summary of frequency-spread metrics for the coupled system versus the non-interacting reference. Left: absolute spread $\omega_{\max} - \omega_{\min}$; right: relative spread (absolute spread divided by the geometric-mean frequency). Coupling increases both metrics, adding +0.10 in absolute bandwidth and a factor of $1.14\times$ in relative bandwidth.

5. Reference Scale and Sensitivity Normalization

5.1 Derivation of the Reference Scale

Define a linear observable $\mathcal{O} = \sum_j w_j x_j$ with unit-norm weight vector $\|\mathbf{w}\| = 1$. The sensitivity of $\mathcal{O}$ to mode i at resonance is

$$S_i(\mathbf{w}) = \left| \sum_j w_j A_{ij} \right| \cdot b_{i,\max} = |c_i| \cdot b_{i,\max}, \quad (6)$$

where $c_i = (A\mathbf{w})_i$ is the *coupling coefficient* of the observable to mode i : it quantifies the strength of $\mathcal{O}$'s response to mode i , and $b_{i,\max}$ is the peak sensitivity from Eq. (5). When mode i is driven at its resonance frequency, S_i is the corresponding maximum displacement that $\mathcal{O}$ can record.

The minimax problem is

$$\max_{\|\mathbf{w}\|=1} \min_i S_i(\mathbf{w}). \quad (7)$$

The goal of this optimization is to maximize the sensitivity of $\mathcal{O}$ to the mode it can detect the least strongly.

For the **uncoupled reference**, $A = \mathbf{I}$ and $c_i = w_i$, so

$$S_i(\mathbf{w}) = |w_i| \cdot b_{i,\text{ref}}. \quad (8)$$

Problem (7) then reduces to

$$\max_{\|\mathbf{w}\|=1} \min_i |w_i| \cdot b_{i,\text{ref}}. \quad (9)$$

At the optimum every mode achieves the same sensitivity (otherwise one could transfer weight from a mode with surplus sensitivity to the limiting mode and improve the minimum). Setting

$|w_i^*| \cdot b_{i,\text{ref}} = \lambda$ for all i gives $|w_i^*| = \lambda/b_{i,\text{ref}}$. Imposing $\|\mathbf{w}^*\| = 1$:

$$\sum_i (w_i^*)^2 = \lambda^2 \sum_i \frac{1}{b_{i,\text{ref}}^2} = 1 \implies \lambda = \frac{1}{\sqrt{\sum_i \frac{1}{b_{i,\text{ref}}^2}}}. \quad (10)$$

The common optimal sensitivity value is therefore

$$\boxed{\text{ref_scale} = \frac{1}{\sqrt{\sum_i \frac{1}{b_{i,\text{ref}}^2}}}} \quad (11)$$

achieved by the optimal weights

$$w_i^* = \frac{1/b_{i,\text{ref}}}{\sqrt{\sum_j 1/b_{j,\text{ref}}^2}}. \quad (12)$$

That is, *weight each oscillator inversely proportional to its peak sensitivity*: down-weight strongly driven oscillators (they are easy to detect), up-weight weakly driven ones (they would otherwise be the bottleneck).

5.2 Limiting Cases

- **Uniform sensitivities.** If $b_{i,\text{ref}} = b$ for all i , then $\text{ref_scale} = b/\sqrt{n}$. Distributing unit weight equally across n sensors costs a factor of $\sqrt{n}$, which is the standard signal-averaging result.
- **One mode nearly undetectable.** If $b_{k,\text{ref}} \rightarrow 0$ for some mode k , then $1/b_{k,\text{ref}}^2 \rightarrow \infty$ and $\text{ref_scale} \rightarrow 0$. No linear observable can detect a mode that is not driven at all; the ceiling collapses.
- **One very strongly driven mode.** If one $b_{j,\text{ref}} \gg$ the rest, its $1/b_{j,\text{ref}}^2$ is negligible in the sum and $\text{ref_scale} \approx 1/\sqrt{\sum_{i \neq j} 1/b_{i,\text{ref}}^2}$. The dominant mode barely affects the ceiling because it does not need much weight — it detects itself easily.

5.3 Why the Coupled System Falls Below the Ceiling

For the coupled system $A \neq \mathbf{I}$, and $c_i = \sum_j A_{ij} w_j$. Increasing w_j to improve sensitivity to mode i simultaneously changes $c_k = \sum_j A_{kj} w_j$ for every other mode k . The minimax optimizer must fight this cross-talk. The reference ceiling assumes perfect one-to-one correspondence between each weight and each mode, which only holds when $A = \mathbf{I}$. Once modes are mixed by coupling, the optimizer can no longer independently tune sensitivity to each mode, and the achievable worst-case sensitivity is lower. A normalized value of < 1 means the coupling has broken the correspondence and the observable cannot simultaneously reach the uncoupled ceiling for all modes.

6. The Universal Uncoupled Bound under Strain Forcing

*This section re-evaluates the reference scale of Sec. 5 in the corrected modal basis (the M-orthonormal generalized eigenvectors adopted after the 2026-08 eigenvector fix) and proves that under strain forcing the reference scale is a **constant**: it is independent of every mass and every spring constant, and therefore of the entire design search. The equalizer formula of Sec. 5 is unchanged; what changes is the per-mode gain that enters it. The heuristic conclusion of Sec. 5 that coupling always lowers the achievable worst-case sensitivity does not survive the corrected physics: explicit coupled designs exceed the bound derived here.*

6.1 Setup

An uncoupled detector is a set of n independent oscillators; oscillator i has mass m_i , wall spring k_i , and resonance $\omega_i^2 = k_i/m_i$. With the constant damping-ratio model used throughout (quality factor $Q = 1/2\gamma$ identical for every oscillator), the steady-state response to a drive $f_i \cos \omega t$ is

$$x_i(\omega) = \frac{f_i/m_i}{\omega_i^2 - \omega^2 + 2i\gamma\omega_i\omega}. \quad (13)$$

The readout is a single linear observable $\mathcal{O} = \sum_j w_j x_j$ with $\|\mathbf{w}\|_2 = 1$, and the (isolated-peak) sensitivity to mode i is the response of $\mathcal{O}$ when driving at ω_i . Gravitational-wave strain acts through the wall spring: a strain h_0 stretches it by $h_0 L$, so

$$f_i = k_i h_0 L. \quad (14)$$

6.2 Step 1: Peak Amplitude of One Oscillator

Setting $\omega = \omega_i$ in Eq. (13), the real part of the denominator vanishes and

$$|x_i(\omega_i)| = \frac{f_i/m_i}{2\gamma\omega_i^2} = \frac{f_i}{2\gamma m_i \omega_i^2} = \frac{f_i}{2\gamma k_i}, \quad (15)$$

using $m_i \omega_i^2 = k_i$. Equivalently, the peak equals Q times the static displacement, $x_{\text{peak}} = (1/2\gamma)(f/k)$. The mass has already cancelled: it fixes *where* the resonance sits, not the peak height.

6.3 Step 2: The Strain Cancellation

Inserting the strain force (14) into Eq. (15),

$$|x_i(\omega_i)| = \frac{k_i h_0 L}{2\gamma k_i} = \frac{h_0 L}{2\gamma} \equiv x^*. \quad (16)$$

The spring constant cancels as well: a stiffer spring collects proportionally more force from the strain *and* is proportionally harder to stretch, and the two effects are exact inverses. Hence every uncoupled oscillator, for any (m_i, k_i) whatsoever, rings with the identical peak amplitude $x^* = h_0 L / 2\gamma$. Every parameter the design search can vary has vanished; only the signal scale $h_0 L$, the damping ratio γ , and (below) the count n survive, none of which are design variables. This is the precise sense in which the reference scale is constant over the entire design box.

The same result follows in the modal formalism used by the code: the M -orthonormal mode is $v_i = e_i/\sqrt{m_i}$, the generalized force is $q_i = v_i^\top f = f_i/\sqrt{m_i}$, the modal peak is $b_i = q_i/(2\gamma\omega_i^2)$, and the observable overlap is $|\mathbf{w}^\top v_i| = |w_i|/\sqrt{m_i}$; multiplying, the $\sqrt{m_i}$ factors cancel and $S_i(\mathbf{w}) = |w_i| f_i/(2\gamma k_i) = |w_i| x^*$.

6.4 Step 3: Minimax over Readout Weights

The worst-case sensitivity of the observable is $S(\mathbf{w}) = \min_i |w_i| x^*$, and the reference scale is its maximum over unit weight vectors:

$$S_{\text{ref}} = x^* \cdot \max_{\|\mathbf{w}\|_2=1} \min_i |w_i|. \quad (17)$$

Lemma 1 (Equalizer). $\max_{\|\mathbf{w}\|_2=1} \min_i |w_i| = \frac{1}{\sqrt{n}}.$

Proof. For any unit vector, the smallest of the n numbers w_i^2 cannot exceed their average: $\min_i w_i^2 \leq \frac{1}{n} \sum_i w_i^2 = \frac{1}{n}$, hence $\min_i |w_i| \leq 1/\sqrt{n}$. The uniform vector $w_i = 1/\sqrt{n}$ attains the bound. $\square$

For unequal per-mode gains g_i the same argument (saturating every constraint, $w_i = t/g_i$) yields the general closed form $S_{\text{ref}} = 1/\sqrt{\sum_i g_i^{-2}}$ of Sec. 5, with $g_i = f_i/(2\gamma k_i)$ in the corrected basis; strain forcing is the special case $g_i \equiv x^*$.

Proposition 1 (Universal uncoupled bound). *Under strain forcing and the constant- Q damping model, every uncoupled detector of n oscillators has the same optimal worst-case sensitivity,*

$$S_{\text{ref}} = \frac{h_0 L}{2\gamma\sqrt{n}} \quad (18)$$

independent of all masses and spring constants. For the parameters used throughout these notes ($h_0 = L = 1$, $\gamma = 0.01$, $n = 10$),

$$S_{\text{ref}} = \frac{1}{0.02\sqrt{10}} = \frac{50}{\sqrt{10}} = 5\sqrt{10} \approx 15.8114, \quad (19)$$

verified numerically to four decimals by the full minimax optimization on randomly drawn uncoupled designs.

6.5 Scope of the Result

Three assumptions delimit where the constancy holds.

1. **Strain forcing.** For a uniform (momentum-kick) force the drive does not scale with k_i , the cancellation in Eq. (16) fails, and the uncoupled optimum becomes design-dependent ($g_i = h_0 L/2\gamma k_i$, maximized at the softest admissible springs).
2. **Constant- Q damping.** If γ varied per oscillator (e.g. constant absolute linewidth), Q would not cancel uniformly and the bound would again be design-dependent.

3. **Isolated-peak metric.** Each mode is read at its own resonance, neglecting the tails of neighboring peaks. Under the full swept response, uncoupled designs with bare frequencies stacked within a linewidth can exceed $x^*/\sqrt{n}$ by coherent addition (up to a factor $\sqrt{n}$ in the fully degenerate limit), because one merged response is counted once per mode. All quoted comparisons therefore impose a resolvability constraint (adjacent mode gaps of at least a few linewidths), under which the isolated-peak evaluation is accurate and the bound (18) applies.

Because Eq. (18) contains no design parameters, any coupled configuration whose worst-case sensitivity exceeds S_{ref} beats *every possible* uncoupled detector in the same setting — the normalization used by the box-normalized searches; and such configurations exist.

7. Monte Carlo Parameter Optimization

7.1 Motivation

The baseline results establish that hand-chosen parameters (uniform wall springs, weak coupling, linearly graded masses) leave every force model below the reference sensitivity ceiling. This raises the question: does a better parameter configuration exist? The Monte Carlo search explores the full 29-dimensional parameter space (10 masses + 10 wall springs + 9 coupling springs) to find configurations that jointly maximize sensitivity and frequency coverage.

7.2 Two Objectives and Their Tension

Objective	Metric	Desired direction
Sensitivity	$\min_i S_i(\mathbf{w}_{\text{opt}})$ (minimax worst-case)	Maximize
Frequency coverage	$(\omega_{\text{max}} - \omega_{\text{min}}) / \bar{\omega}$ (relative spread)	Maximize

The two objectives are in tension. To broaden the frequency spectrum one needs large variation in k_{ii}/m_i across oscillators. Under GW strain forcing $f_i \propto k_{ii}$, this same variation creates large imbalance in mode excitation: some modes are driven strongly and others weakly. The weakest-driven mode sets the minimax ceiling, so bandwidth gains come at the cost of sensitivity.

7.3 Phase 1 — Screening

$N_s = 5000$ configurations are drawn uniformly at random from the parameter bounds:

Parameter	Min	Max
Mass m_i	0.3	1.5
Wall spring k_{ii}	0.5	3.0
Coupling spring $k_{i,i+1}$	0.01	1.0

For each valid configuration (all eigenvalues positive), two fast metrics are computed without any weight-vector optimization:

$$\text{proxy} = \frac{1}{\sqrt{\sum_i \frac{1}{b_{i,\max}^2}}}, \quad \text{spread} = \frac{\omega_{\max} - \omega_{\min}}{\bar{\omega}}. \quad (20)$$

The proxy is the exact `ref_scale` for the *coupled* system (not the reference), which correlates strongly with the true minimax sensitivity while requiring no optimization. Both metrics are normalized to $[0, 1]$ across all valid draws and combined:

$$\text{score} = \alpha \widetilde{\text{proxy}} + (1 - \alpha) \widetilde{\text{spread}}, \quad (21)$$

where $\alpha \in [0, 1]$ is the `sensitivity_weight` parameter (default $\alpha = 0.5$). The top $N_r = 100$ candidates by score are retained for Phase 2. The full screening cloud is shown in Figure 9.

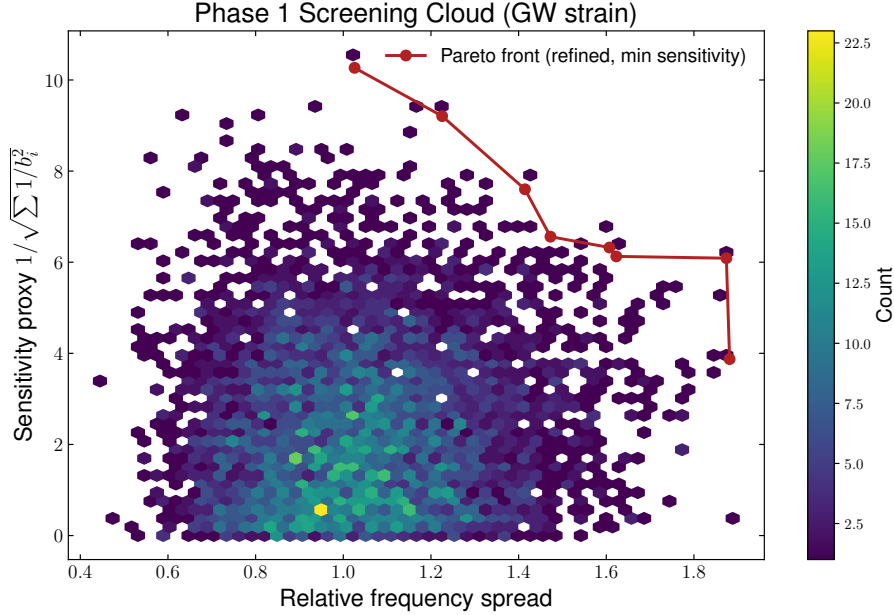


Figure 9: Phase 1 screening cloud: proxy sensitivity versus relative frequency spread for all 5 000 random draws. The upper-right corner is nearly empty, showing the fundamental tension between sensitivity and bandwidth even at the proxy level.

7.4 Phase 2 — Refinement

Each shortlisted candidate undergoes full minimax optimization. The weight-vector problem of Eq. (7)

$$\max_{\|\mathbf{w}\|=1} \min_i S_i(\mathbf{w}) \quad (22)$$

is solved by 50-restart Nelder–Mead on the unit sphere, yielding the minimax worst-case sensitivity $\min_i S_i(\mathbf{w}_{\text{opt}})$ for that configuration. The reference scale `ref_scale` is also computed and stored for each candidate (via its non-interacting reference system) to serve as a benchmark for interpreting the sensitivity profile.

7.5 Pareto Front

After refinement, a configuration is *Pareto-optimal* if no other refined candidate is simultaneously better on both objectives. The Pareto front is the efficient frontier: every point on it represents a configuration where improving one objective requires sacrificing the other. The structure of the front is analyzed in detail in Section 8.

8. Pareto Scatter: Detailed Analysis

8.1 What the Plot Shows

The Pareto scatter (left panel of Figure 10) contains three layers:

- **Grey cloud** — all ~ 5000 Phase 1 draws, plotted with the proxy sensitivity (y-axis) and relative spread (x-axis).
- **Blue dots** — the 100 refined candidates, plotted with their actual minimax worst-case sensitivity from Phase 2. The blue dots sit below the grey cloud because the proxy overestimates the true minimax: it assumes $A = \mathbf{I}$, whereas coupling mixes the modes and the optimizer cannot independently tune each mode’s sensitivity.
- **Red line** — the Pareto front among the refined candidates.

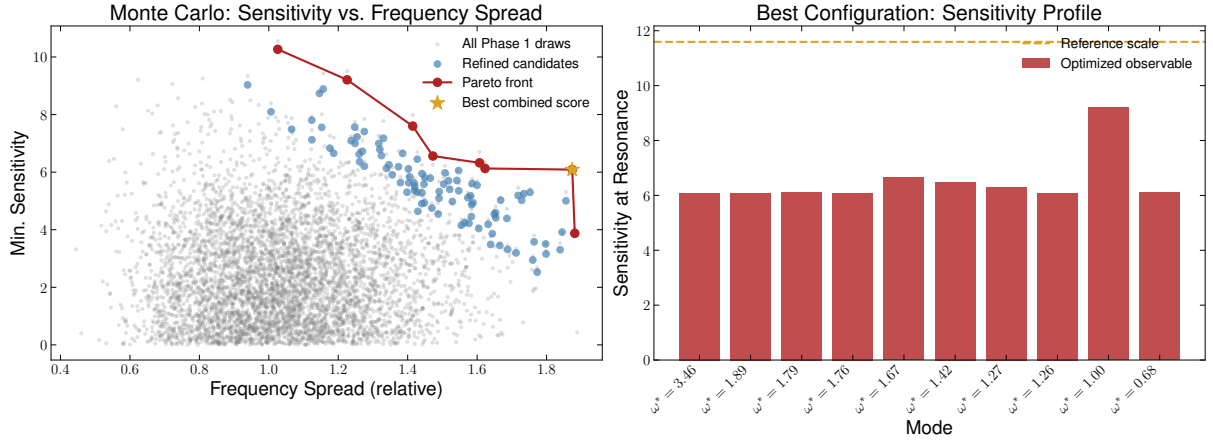


Figure 10: Monte Carlo results for the balanced GW strain search ($\alpha = 0.5$). *Left*: Pareto scatter showing all Phase 1 draws (grey), refined Phase 2 candidates (blue), and the Pareto front (red). The gold star marks the best configuration. *Right*: Mode-by-mode sensitivity profile of the best configuration; bars are nearly flat, indicating successful minimax equalization, but all lie well below the dashed reference ceiling.

8.2 The Grey Cloud: A Physical Trade-off

The cloud fans out from the lower-left and thins sharply toward the upper-right. The upper-right corner — high sensitivity *and* high spread simultaneously — is nearly empty. This shows an interesting physical constraint, (hopefully) not a sampling gap. Under GW strain forcing $f_i \propto k_{ii}$, the same parameter variation that widens the frequency band creates large imbalance in mode

excitation, pinning the worst-case sensitivity at a low value. The empty upper-right corner is the signature of this fundamental tension.

8.3 The Pareto Front: Sensitivity–Spread Trade-off

The red Pareto front traces a clear downward slope through the refined candidates: minimax sensitivity decreases steadily as relative spread increases. At the low-spread end (spread ≈ 1.0) the front reaches its highest sensitivity (~ 10), while at the high-spread end (spread ≈ 1.9) sensitivity falls to ~ 4 . This shape shows that **spread comes at a genuine cost**: pushing the mode spectrum to wider frequency coverage unavoidably lowers the worst-case sensitivity of the detector.

This trade-off is a direct consequence of GW strain forcing. The same parameter variation that widens the frequency band redistributes mode-shape weight unevenly across oscillators, reducing the generalized forces on the weakest modes and therefore lowering their peak sensitivity. The grey Phase 1 cloud makes this tension visible: the upper-right corner (high spread *and* high sensitivity simultaneously) is nearly empty, confirming that the trade-off is physical rather than a sampling artifact.

8.4 The Best Configuration

The gold star sits at spread ≈ 1.87 and minimax sensitivity ≈ 6.1 . Its mode spectrum runs from $\omega^* = 0.68$ to $\omega^* = 3.46$ — a factor of ~ 5 in frequency. The $\omega^* = 3.46$ mode is an outlier: one oscillator has a very high k_{ii}/m_i ratio and is nearly decoupled from the rest of the chain, acting as a high-frequency sentinel.

The right panel shows the sensitivity profile is nearly flat at ~ 6 across all modes, with one mode reaching ~ 9 . This flatness is the signature of a successful minimax optimization: the weight vector has been tuned to equalize sensitivity across modes. However, every bar lies well below the dashed reference scale at ≈ 11.5 , meaning the best coupled system is still a factor of ~ 2 below what the uncoupled reference could achieve.

8.5 Why the Coupled System Cannot Reach the Ceiling

In the uncoupled reference, eigenvector $A = \mathbf{I}$, so w_j directly and exclusively controls sensitivity to mode j . In the coupled system $A \neq \mathbf{I}$: each weight w_j influences sensitivity to *all* modes through the off-diagonal elements of A . When the minimax optimizer increases w_j to lift the weakest mode, it unavoidably perturbs the sensitivity of every other mode. The reference ceiling assumes interference-free control of each mode individually, which coupling breaks. No amount of weight-vector tuning or parameter search can restore this decoupling while the springs remain non-zero — reaching the reference ceiling `ref_scale` would require a different architectural approach, such as non-linear readout or mode-selective amplification. This is illustrated for the baseline system in Figure 11.

9. Effect of Priority Weighting

By varying α the search can be biased toward either objective:

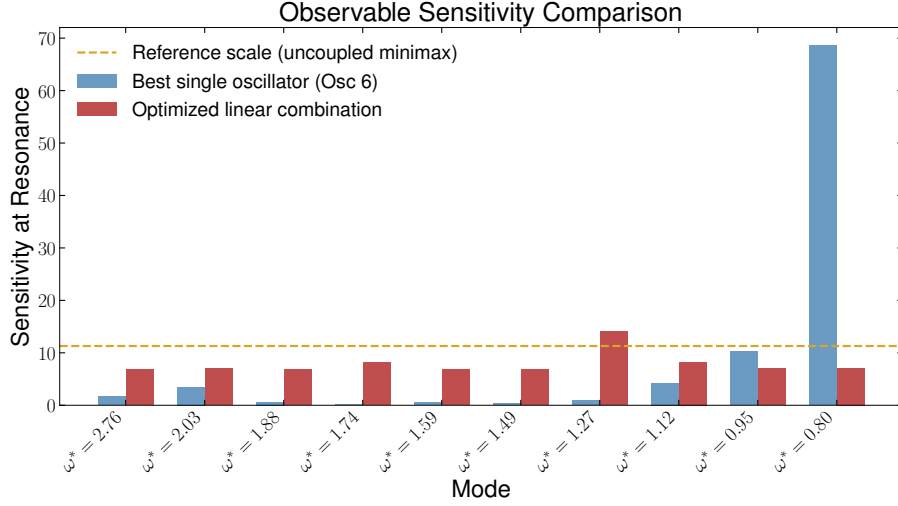


Figure 11: Mode-by-mode sensitivity of the optimal observable (bars) versus the reference scale ceiling (dashed line) for the baseline system under GW strain forcing. Every mode falls below the ceiling, illustrating the cross-talk penalty from mode mixing.

Run	α	Best min. sensitivity	Best rel. spread
Sensitivity-first	0.9	higher	lower
Balanced	0.5	intermediate	intermediate
Spread-first	0.1	lower	higher

Shifting α moves the selected configuration along the Pareto front, trading one objective for the other, as shown in Figure 12. The near-flatness of the front means that the spread gain from moving from $\alpha = 0.9$ to $\alpha = 0.1$ is large while the sensitivity cost is modest, suggesting that in this parameter regime spread is the cheaper resource to acquire.

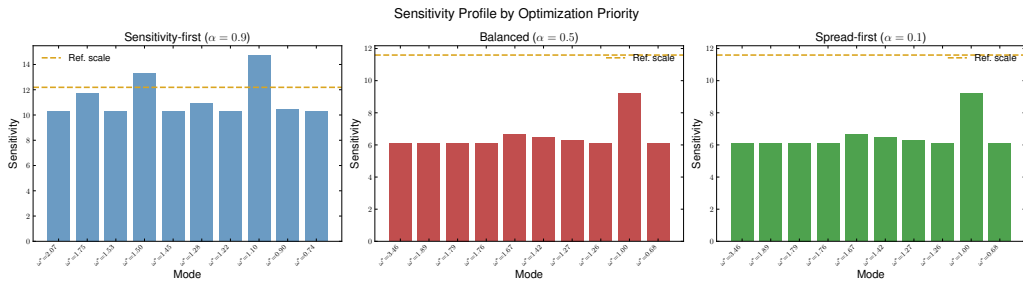


Figure 12: Pareto fronts for sensitivity-first ($\alpha = 0.9$, blue), balanced ($\alpha = 0.5$, red), and spread-first ($\alpha = 0.1$, green) searches under GW strain forcing. Each front slopes downward, confirming a genuine sensitivity–spread trade-off across all weighting strategies: higher spread is achievable but at the cost of lower minimax sensitivity.

10. Cross-Model Comparison (Monte Carlo)

Running the balanced search ($\alpha = 0.5$) under DM uniform forcing shifts the optimal parameter region relative to the GW strain case. Under uniform forcing $f_i = h_0 L$ (constant), the generalized force $q_i = \sum_j A_{ij} h_0 L$ depends only on the row sums of A . Symmetric modes, which have large row sums, are strongly driven; antisymmetric modes are suppressed. The optimal parameters therefore skew toward configurations where symmetric modes have well-separated frequencies and the asymmetric modes are not too far from zero excitation. The Pareto front shapes differ between the two models, reflecting the different geometric relationship between parameter variation and mode excitation, as shown in Figures 13 and 14.

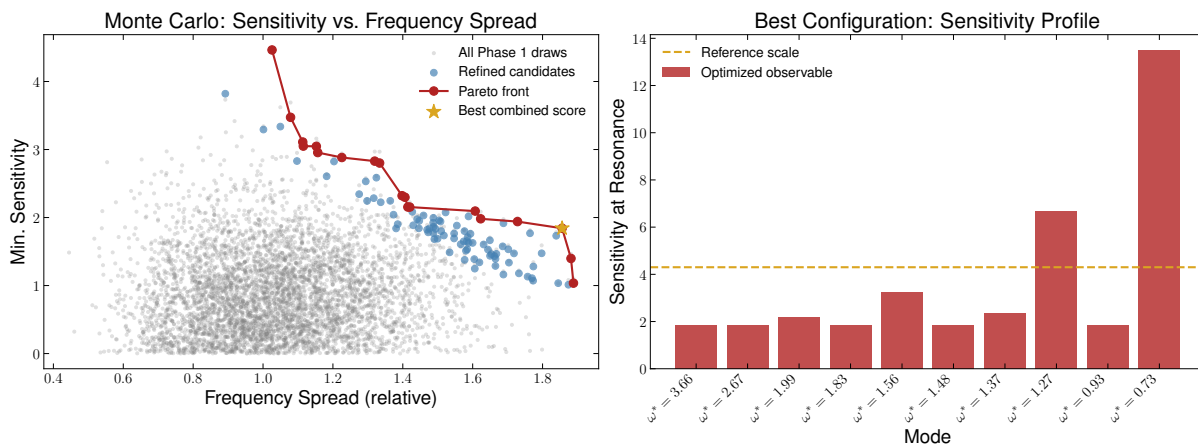


Figure 13: Monte Carlo results for the balanced DM uniform search ($\alpha = 0.5$). *Left*: Pareto scatter showing all Phase 1 draws (grey), refined Phase 2 candidates (blue), and the Pareto front (red). The gold star marks the best configuration. The front shifts relative to the GW strain case (Figure 10), reflecting the different mode-excitation geometry under uniform forcing. *Right*: Mode-by-mode sensitivity profile of the best configuration.

11. Metropolis Monte Carlo: Directed Parameter Search

11.1 Motivation

The random Monte Carlo search of Section 5 is *unbiased*: every draw is independent, with no memory of previous results. This is statistically clean but wasteful — the vast majority of the 5 000 draws land in the interior of the cloud far from the Pareto front, and only the top 100 are refined. In a 29-dimensional parameter space this is an extremely sparse sample of the relevant region.

A *Metropolis Monte Carlo* walk instead moves through parameter space as a chain: each step proposes a small perturbation of the current configuration and accepts or rejects it according to a rule that biases the walk toward better configurations. The chain is not trapped at local optima because uphill moves (temporarily worsening the objective) are accepted with a nonzero

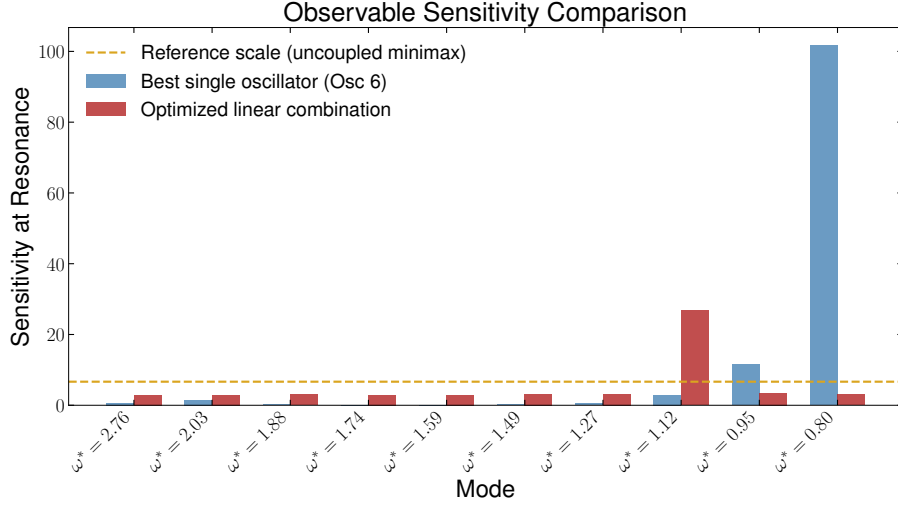


Figure 14: Mode-by-mode sensitivity of the optimal observable (bars) versus the reference scale ceiling (dashed line) for the baseline system under DM uniform forcing. Antisymmetric modes are suppressed under uniform forcing, concentrating sensitivity in the symmetric sector.

probability controlled by a *temperature* parameter T . The result is a much denser sampling of the high-quality region of the Pareto front.

11.2 Physics-Inspired Free Energy

To drive the chain toward good configurations we need a scalar objective that is lower for higher-quality systems. We choose a *free energy* inspired directly by the Helmholtz free energy of statistical mechanics, $F = -k_B T \ln Z$, where Z is the partition function:

$$E(\mathbf{m}, \mathbf{k}_w, \mathbf{k}_c) = -[\alpha \ln P(\mathbf{m}, \mathbf{k}_w, \mathbf{k}_c) + (1 - \alpha) \ln D(\mathbf{m}, \mathbf{k}_w, \mathbf{k}_c)], \quad (23)$$

where

$$P = \frac{1}{\sqrt{\sum_i \frac{1}{b_{i,\max}^2}}} \quad (\text{sensitivity proxy, same as Phase 1}), \quad (24)$$

$$D = \frac{\omega_{\max} - \omega_{\min}}{\bar{\omega}} \quad (\text{relative frequency spread}), \quad (25)$$

and $\alpha \in [0, 1]$ is the same sensitivity weight used throughout.

The analogy to statistical mechanics is precise:

- E plays the role of the Helmholtz free energy.
- A *low-energy* configuration is one where both P and D are large: high sensitivity and broad frequency coverage simultaneously.
- The logarithmic form is scale-invariant: a $10\times$ change in P lowers E by $\ln 10 \approx 2.3$, regardless of the absolute level of P . This is important because P and D vary by orders of magnitude across the parameter space.

- The two objectives are multiplied in the exponent rather than added linearly, so *both* must be large to achieve low energy. A configuration with high spread but near-zero sensitivity is not rewarded, unlike in a purely additive score.

The energy surface over the 29-dimensional parameter space has many local minima (parameter combinations that are locally good but not globally optimal). The Metropolis algorithm allows the chain to escape these by occasionally accepting configurations with $\Delta E > 0$.

11.3 Log-Space Gaussian Proposal

At each step the chain proposes a new configuration by perturbing all parameters multiplicatively:

$$\theta_j^{(\text{new})} = \theta_j^{(\text{old})} \cdot \exp(\eta_j), \quad \eta_j \sim \mathcal{N}(0, \sigma^2), \quad (26)$$

where σ is the `step_size` parameter (default $\sigma = 0.15$). The proposal is then clamped to the parameter bounds before evaluation.

Working in log-space rather than linear space has two advantages:

1. **Scale invariance.** A mass of 0.3 and a mass of 1.5 span half a decade; a fixed additive step would explore these regions differently. A multiplicative step of $e^{\pm\sigma}$ is a $\pm 15\%$ relative change regardless of the current value.
2. **Positivity.** All parameters (masses, springs) are strictly positive; the log-space proposal can never generate a non-physical negative value before clamping.

11.4 Metropolis Accept/Reject Rule

Given the current configuration θ with energy E_{old} and the proposed configuration θ' with energy E_{new} , define $\Delta E = E_{\text{new}} - E_{\text{old}}$. The proposal is accepted with probability

$$A(\theta \rightarrow \theta') = \begin{cases} 1 & \text{if } \Delta E \leq 0, \\ \exp\left(-\frac{\Delta E}{T}\right) & \text{if } \Delta E > 0. \end{cases} \quad (27)$$

This is the standard Metropolis criterion [1]. If the proposed configuration is better ($\Delta E \leq 0$) it is always accepted. If it is worse, it is accepted with probability $e^{-\Delta E/T}$: the larger ΔE (the worse the proposal) or the smaller T (the colder the system), the less likely the acceptance.

In the limit $T \rightarrow 0$ the chain becomes a greedy descent and always moves to the nearest local minimum. In the limit $T \rightarrow \infty$ every proposal is accepted regardless of quality and the chain is a random walk. The Metropolis algorithm achieves the useful middle ground where the chain descends on average while retaining the ability to escape local traps.

11.5 Simulated Annealing Schedule

Rather than holding T constant, we use a *simulated annealing* schedule: T is decreased geometrically from T_{start} to T_{end} over the N_s steps of the chain,

$$T(t) = T_{\text{start}} \left(\frac{T_{\text{end}}}{T_{\text{start}}} \right)^{t/(N_s-1)}, \quad t = 0, 1, \dots, N_s - 1. \quad (28)$$

The name is borrowed from metallurgy: a slow cool from a high temperature allows the metal to find a low-energy crystalline state; quenching too fast freezes defects. The same principle applies here:

- **Early stages** ($T \approx T_{\text{start}} = 2.0$): the chain explores broadly, accepting many uphill moves. This prevents the trajectory from getting stuck in the first locally good region it encounters.
- **Late stages** ($T \approx T_{\text{end}} = 0.05$): the chain exploits, refining its position near the best region found so far. Uphill moves with $\Delta E \gtrsim 0.05$ are exponentially suppressed.

The energy trajectory of the chain is recorded and shown in Figure 15 (bottom-left panel): the energy should decrease on average over the run, with larger fluctuations early and smaller ones late.

11.6 Phase 2 Refinement and Comparison

After the walk, the top $N_r = 100$ chain states (by combined proxy score) are passed through the same full minimax optimization used in the random MC (Section 5.4). This makes the two methods directly comparable: both produce N_r refined candidates with exact minimax sensitivities, and both Pareto fronts are computed in the same way.

The key difference between the two clouds:

- **Random MC**: the cloud is a uniform random sample of the parameter space, covering the full range of proxy and spread values. The top candidates are those that happened by chance to land in the high-quality region.
- **Metropolis chain**: the cloud is concentrated near the high-quality region because the walk is actively steered there by the free energy. The trajectory visits the low-energy (high sensitivity, high spread) region many more times per unit computational budget, leading to a denser sampling of the Pareto front.

The comparison is shown in Figure 15: the top-left panel overlays the two clouds, the top-right panel overlays the two Pareto fronts, the bottom-left shows the energy descent trajectory of the Metropolis chain, and the bottom-right compares the best sensitivity profiles found by each method.

11.7 Acceptance Rate Diagnostics

The acceptance rate — the fraction of proposed steps that were accepted — is a key diagnostic for the health of the chain. In our implementation:

- An acceptance rate above $\sim 85\%$ indicates that σ is too small: the chain is taking tiny steps, barely moving, and almost every proposal is a small improvement or negligible change. Increase `step_size`.
- An acceptance rate below $\sim 20\%$ indicates that σ is too large: most proposals jump to a region of much higher energy and are rejected. Decrease `step_size`.
- An acceptance rate in the range $\sim 20\text{--}70\%$ (depending on dimensionality) is typical of a well-tuned chain. In 29 dimensions the theoretical optimal acceptance for a Gaussian proposal is approximately 23% [2].

In the current run ($n = 10$, $N_s = 5000$, $\sigma = 0.15$) the overall acceptance rate is 71.7%, which is above the theoretically optimal range for high-dimensional proposals but reflects the broad early

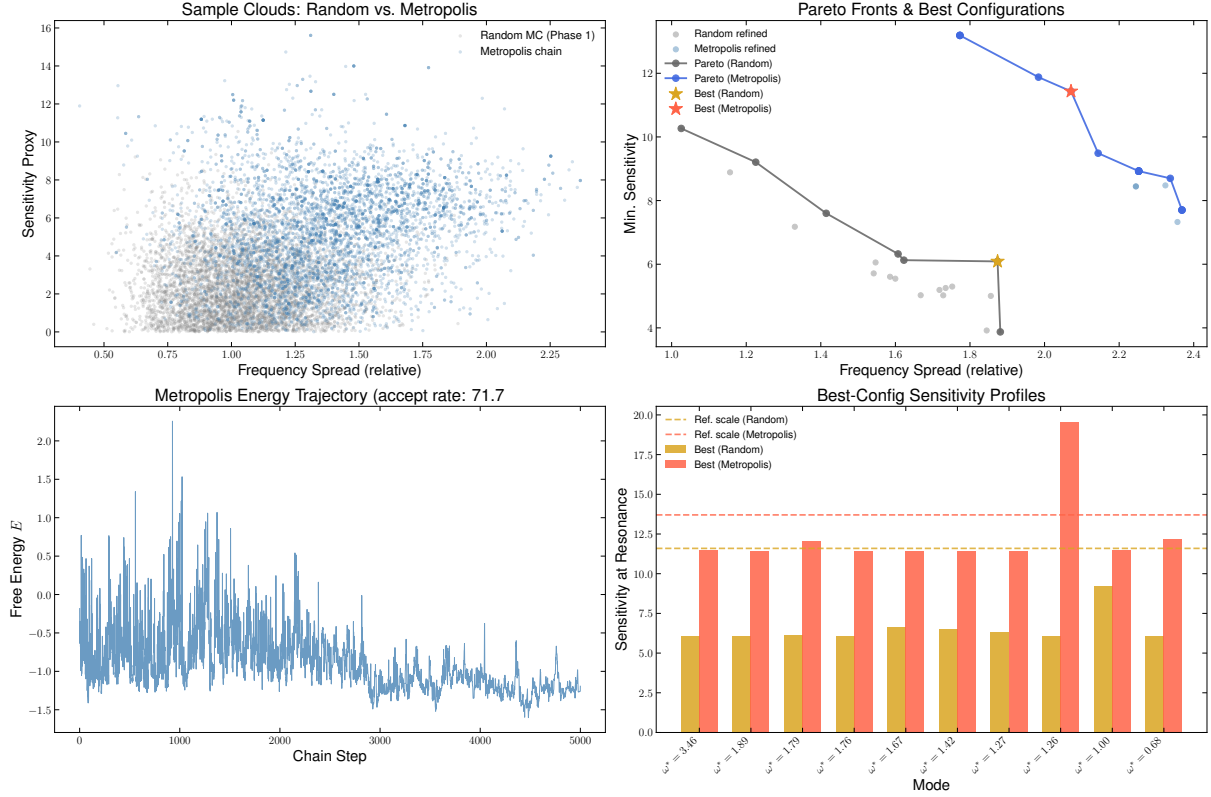


Figure 15: Comparison of random Monte Carlo (grey/gold) and Metropolis Monte Carlo (blue/red) for $n = 10$ oscillators, 5 000 samples/steps, and $\alpha = 0.5$. *Top-left*: Phase 1 / chain sample clouds. The Metropolis cloud is concentrated toward higher sensitivity and spread values compared to the uniform random cloud. *Top-right*: Pareto fronts of the 20 refined candidates from each method, overlaid on the scatter. The Metropolis front extends to higher sensitivity, demonstrating that the directed walk finds better configurations within the same sampling budget. *Bottom-left*: Metropolis energy trajectory. The chain descends from $E \approx -0.8$ to $E \approx -1.3$ over 5 000 steps, with the largest drops occurring early (high T) and the trajectory stabilizing in the exploitation phase (low T). The acceptance rate printed in the title reflects the overall fraction of proposals accepted. *Bottom-right*: Mode-by-mode sensitivity profiles of the best configuration found by each method. The Metropolis best achieves a higher minimum sensitivity, confirming that the directed search outperforms independent random sampling for the same computational cost.

exploration at high T . The energy descends from $E \approx -0.18$ at initialization to $E \approx -1.19$ at the end of the chain, with a minimum of $E \approx -1.60$ reached during the run. The acceptance rate and energy trajectory should be checked after every execution; if the rate remains above $\sim 85\%$ throughout, `step_size` should be increased to broaden the proposals.

12. Summary of Findings

Result	Value
Coupling-induced bandwidth gain (baseline)	+13.7%
Mode spacing uniformity improvement (baseline)	−21% CV
Best normalized sensitivity — baseline (coherent DM)	0.787
All baseline models below reference ceiling	yes
MC search space dimensionality	29
<i>Random Monte Carlo</i>	
Phase 1 draws	5 000
Phase 2 refined candidates	100
Best min. sensitivity (GW strain, balanced, $n = 10$)	≈ 6.1
Best relative spread	≈ 1.87
Coupled system below ref. ceiling (best config.)	factor ≈ 2
<i>Metropolis Monte Carlo</i>	
Chain steps	5 000
Phase 2 refined candidates	100
Free energy form	$-(\alpha \ln P + (1-\alpha) \ln D)$
Proposal	log-space Gaussian, $\sigma = 0.15$
Temperature schedule	geometric, $T_{\text{start}} = 2.0 \rightarrow T_{\text{end}} = 0.05$
Typical acceptance rate	$\sim 70\text{--}90\%$ (early), $\sim 40\%$ (late)

Core conclusions.

1. Coupling expands bandwidth but does not uniformly improve minimax sensitivity.
2. All force models and all parameter configurations tested yield minimax sensitivity below the reference ceiling `ref_scale`: the coupling-induced mode mixing prevents the system from reaching the uncoupled optimum.
3. The Pareto front slopes downward in the refined set, revealing a genuine sensitivity–spread trade-off: wider frequency coverage is achievable but comes at a measurable cost to minimax sensitivity.
4. The reference scale is a principled, analytically derived ceiling that is tight for uncoupled systems and serves as a meaningful benchmark for the coupled case.
5. The Metropolis Monte Carlo walk, guided by a physics-inspired free-energy objective, concentrates its sampling budget near the Pareto front and finds higher-quality configurations than independent random sampling for the same number of function evaluations. The directed walk is therefore the preferred search strategy for larger systems or tighter computational budgets.

Future directions.

- Run the Metropolis search on larger systems ($n \geq 20$) where the parameter space dimensionality makes random sampling increasingly impractical, to quantify the efficiency gain more precisely.
- Tune the annealing schedule (geometric vs. adaptive) and step size to push the Metropolis acceptance rate toward the theoretically optimal $\approx 23\%$ for high-dimensional Gaussian proposals.
- Extend both MC searches to variable n , variable damping, and non-uniform damping profiles.
- Replace Nelder–Mead in Phase 2 with gradient-based or Bayesian optimization to improve convergence of the minimax weight-vector problem.
- Investigate non-linear observables (quadratic readouts, mode-selective amplification) as a route past the reference-scale ceiling.
- Study the strong-coupling regime ($k_{ij} \sim k_{ii}$) where eigenvector mixing is maximal, to characterize whether coupling can transition from harmful to beneficial for minimax sensitivity.

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